

FUNDAMENTALS

In this lesson, we will be going through some fundamentals you need to know as they will be used throughout. This includes:

- **Creating a Project Folder.**
- **Creating and Saving a Python File.**
- **Running your programs.**
- **Comments**

Creating a Project Folder

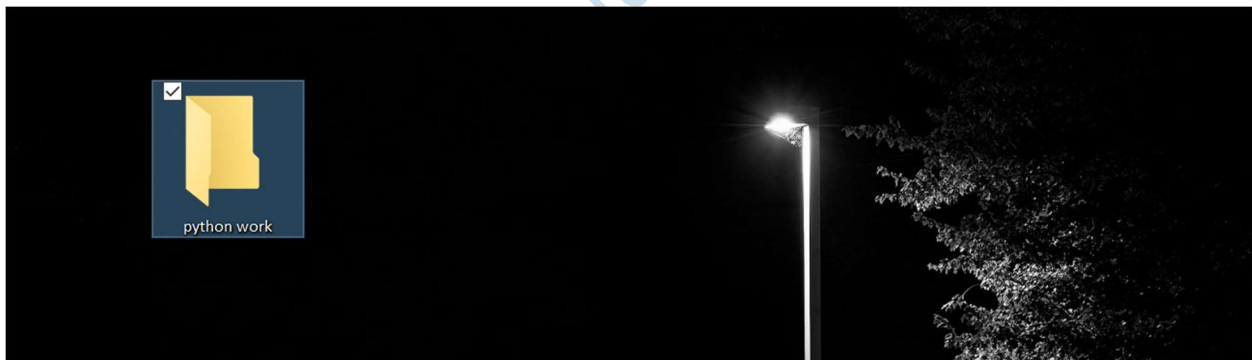
Creating a project folder is an essential habit you need to emulate as a beginner.

A project folder allows you to:

- **Organize your files neatly.**
- **Makes it easy to locate your work**
- **Understand real-world development habits** – Professional software projects are organized into directories. It also allows you to collaborate with other developers easily.

As a beginner, you'll only need one folder for now but note that **different projects require different folders.**

For your first folder, on your **Desktop** press **CTRL + Shift + N** and rename the folder **python work**.



This is the folder where you'll be saving your python files.

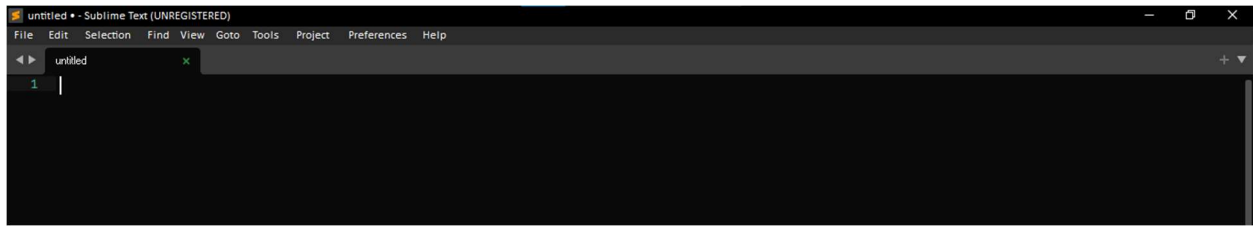
Creating and Saving a Python File

A python file is a file that has instructions written in Python programming language. The files typically end with a **.py** extension so that computers know they contain python code.

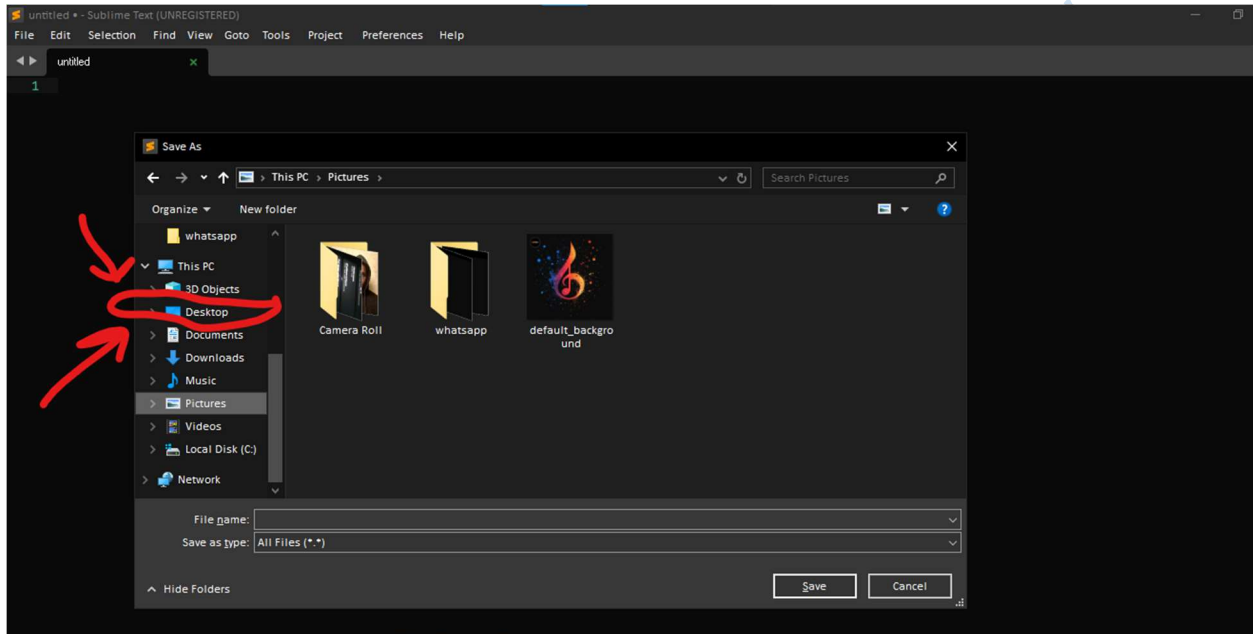
To create a python file, open your code editor and press **CTRL + N**



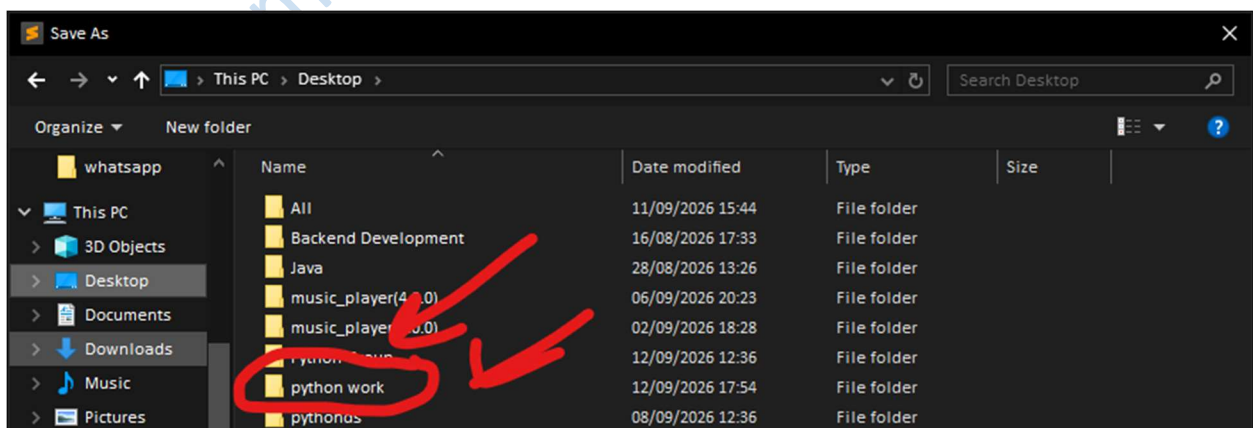
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Now press **CTRL + S** to save this file. A dialog box should pop up.

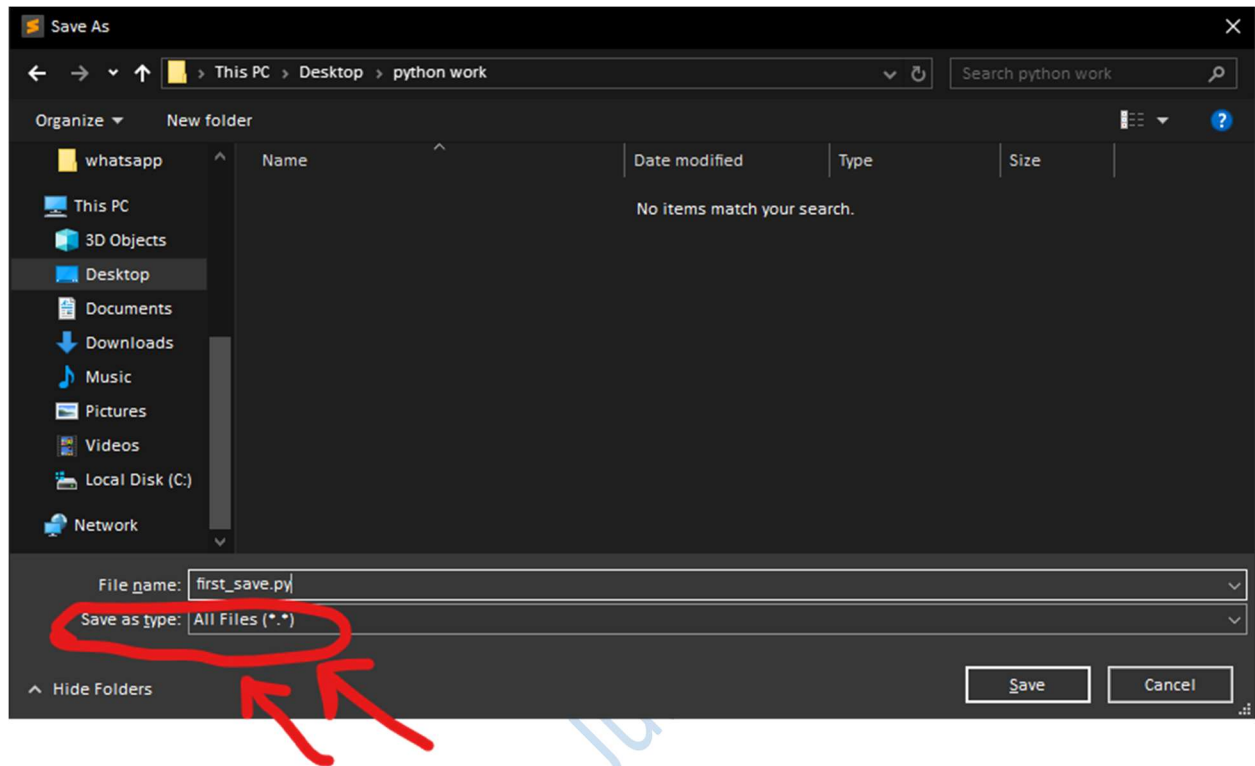


Click Desktop if you created your project folder on your desktop. If not, navigate to where you created your folder.



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Open the folder you created to save the file you are currently choosing the location to save to.



Ensure **Save as type** remains at **All Files**. Then, input your file name as **first_save.py**. Finally, click save.



If you have this, it means you can now create a python file and save it.

Running Your Programs

To run your program, in **Sublime Text**, press **CTRL+B**. For those using **VS Code**, you can press **CTRL+F5**. Let's try this out.

Open **first_save.py** and type this:



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```
C:\Users\njrus\Desktop\python work\first_save.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
first_save.py x
1 print('This is my first save')
```

Run your program using the shortcuts specified above.

```
C:\Users\njrus\Desktop\python work\first_save.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
first_save.py x
1 print('This is my first save')

This is my first save
[Finished in 679ms]
```

This is how we will be running our programs at this level. For more complicated programs, I'll guide on how to use the terminal to run your programs.

Comments

Comments are notes written in a program to provide information about the code. They are ignored by Python when the program is executed. Comments are used to:

- Explain what the code does
- Remind yourself why you wrote something in a particular way
- Make the code easier for other programmers to understand.
- Make a program easier to maintain

Consider the following example:

```
4
5 # Calculate the total price
6
7 total_price = price_per_item * quantity_bought
```

The comment at line 5 explains what the code at line 7 does.

In python, comments are created using the **hash symbol (#)**. Anything written after this symbol is considered a comment and is ignored during program execution.

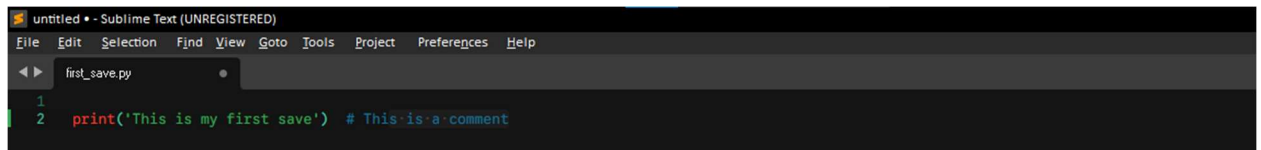
➔ Single-line comments

A single-line comment begins with a **#** and continues until the end of that line. These comments, can also be placed after a line of code.

```
first_save.py
1
2 # This is a comment
3
4 print('This is my first save')
```



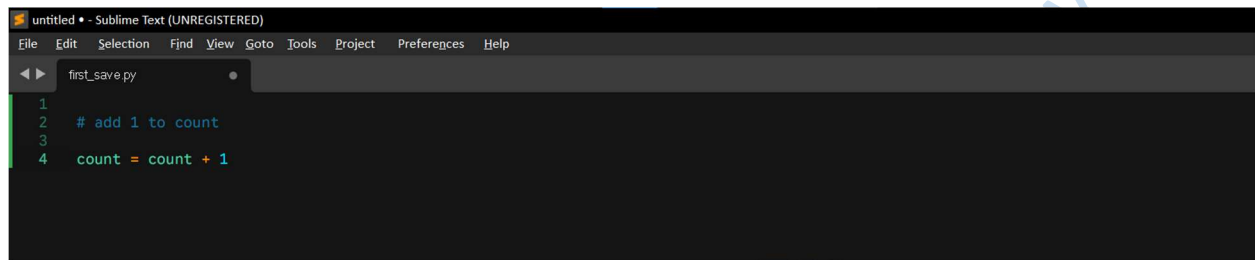
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```
untitled - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
first_save.py
1
2 print('This is my first save') # This is a comment
```

N/B Don't fill your programs with comments just for the sake of having comments. Good code should be understandable on its own where possible.

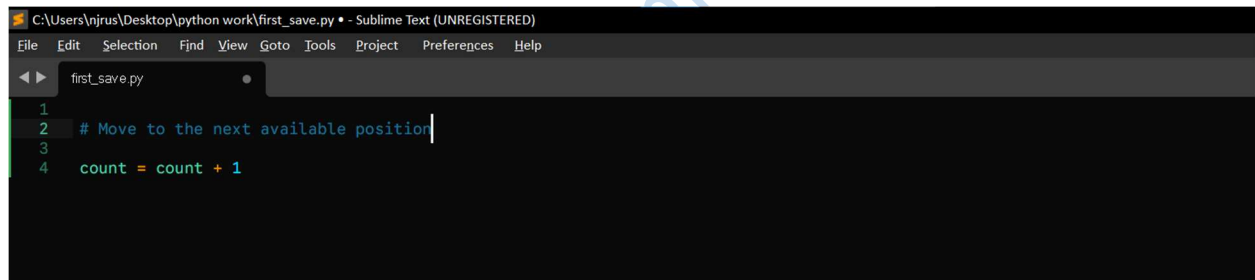
Consider the following program:



```
untitled - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
first_save.py
1
2 # add 1 to count
3
4 count = count + 1
```

The comment on line 2 isn't useful at all since the code on line 4 makes itself obvious.

Now consider this program:



```
C:\Users\njrus\Desktop\python work\first_save.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
first_save.py
1
2 # Move to the next available position
3
4 count = count + 1
```

The comment now explains why we chose to increment the value of count by 1 unlike the earlier one.

Comments are meant to provide information or explain the 'why'. So, decide when to use a comment and when not to.

